

Chapter 8.

Cauchy Goursat Theorem.

Let f be analytic in an open disk Δ . Then, for any closed curve γ in Δ ,

$$\int_{\gamma} f = 0$$

Proof along the Alfors. First, we will show that: for $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$,

$$\int_{\partial R} f = 0, \text{ if } f \text{ is analytic on } R.$$

Denote $\eta(R) \stackrel{\text{def}}{=} \int_{\partial R} f(z) dz$. And, define bisections of R as

$$R^{(1)} = \{(x, y) \mid a \leq x \leq \frac{a+b}{2}, \frac{c+d}{2} \leq y \leq d\} \text{ and}$$

$$R^{(2)} = (\frac{b-a}{2}, 0) + R^{(1)}, R^{(3)} = (0, -\frac{d-c}{2}) + R^{(1)}, R^{(4)} = (\frac{b-a}{2}, -\frac{d-c}{2}) + R^{(1)}.$$

Then $\eta(R) = \eta(R^{(1)}) + \eta(R^{(2)}) + \eta(R^{(3)}) + \eta(R^{(4)})$, so by the triangle inequality,

$$|\eta(R)| \leq \sum_{i=1}^4 \eta(R^{(i)}) \leq \sum_{i=1}^4 \max_{j=1, \dots, 4} \eta(R^{(j)}) = 4 \cdot \eta(R^{(k)}) \text{ for some } k=1, 2, 3, 4$$

Take $R_1 = R^{(k)}$. Repeat this process inductively, we obtain that

$$\{R_n\}_{n=0}^{\infty} \text{ s.t. } R_{n+1} \subseteq R_n \text{ for all } n \geq 0, R_n \neq \emptyset, R_n \text{ compact, and}$$

$$|\eta(R_n)| \geq 4^{-1} |\eta(R_{n-1})| \geq 4^{-2} |\eta(R_{n-2})| \geq \dots \geq 4^{-n} |\eta(R_0)|.$$

Since $\bigcap_{n=0}^{\infty} R_n$ contains exactly one point, denote this z_0 , and since f is analytic,

$$\text{given } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that } |z - z_0| < \delta \Rightarrow \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \epsilon.$$

Since for all $n \geq 0$, R_n contains z_0 , for some $n \in \mathbb{N}$, $R_n \subseteq B_{\delta}(z_0)$. So,

$$\int_{R_n} f(z) dz = \int_{R_n} \underbrace{[f(z) - f(z_0) - f'(z_0) \cdot [z - z_0]]}_{=0} dz \text{ follows that}$$

$$|\eta(R_n)| = \left| \int_{\partial R_n} f(z) dz \right| \leq \epsilon \cdot \int_{\partial R_n} |z - z_0| |dz| \leq \epsilon \cdot \text{diam } R_n \cdot \frac{L}{2^n} \text{ where } L = 2 \cdot \left(\frac{b-a}{2} + \frac{d-c}{2} \right) \\ = \epsilon \cdot \text{diam } R \cdot L \cdot \frac{1}{4^n}.$$

$$\text{so, } |\eta(R)| \leq 4^n \cdot |\eta(R_n)| = \epsilon L \cdot \text{diam } R \rightarrow 0 \text{ as } \epsilon \rightarrow 0.$$

Thm. 9.6. [Riemann's Theorem.]

Suppose that f has a isolated singularity at $z=z_0$ and it is bounded in $\mathbb{B}_R(z_0) \setminus \{z_0\}$ for some $R>0$.

Then there exists a unique analytic extension of f at z_0 .

Proof By the assumption, let f be analytic in $0 < |z-z_0| \leq R$.

Given z_1 lies in the punctured disk, let $r>0$ so that $r < |z_1-z_0| < R$.

Then, by the Cauchy's Thm, for $C: |z-z_0|=R$ and $C_r: |z-z_0|=r$,

$$f(z_1) = \frac{1}{2\pi i} \cdot \left[\int_C \frac{f(u)}{u-z_1} du - \int_{C_r} \frac{f(u)}{u-z_1} du \right]$$

Meanwhile, since f is bounded, set $M = \sup |f|$, so that

$$\begin{aligned} \left| \frac{1}{2\pi i} \int_{C_r} \frac{f(u)}{u-z_1} du \right| &\leq \frac{M}{2\pi} \cdot \int_{C_r} \frac{1}{|u-z_1|} |du| \\ &\leq \frac{M}{2\pi} \cdot \frac{2\pi r}{|z_1-z_0|-r} \rightarrow 0 \text{ as } r \rightarrow 0. \end{aligned}$$

$$\text{so, } f(z_1) = \frac{1}{2\pi i} \int_C \frac{f(u)}{u-z_1} du$$

Since the z_1 was arbitrary,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(u)}{u-z} du, \text{ which is analytic in } |z-z_0| < R.$$